

# CM0832 Elements of Set Theory

## 8. Ordinals and Order Types

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# Preliminaries

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## Textbook

Herbert B. Enderton (1977). Elements of Set Theory.

## Convention

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

# Ordinal Addition

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## Definition

Let  $\alpha$  and  $\beta$  be ordinals. We define their **addition** by transfinite recursion on  $\beta$ :

$$\alpha + 0 = \alpha,$$

$$\alpha + \beta^+ = (\alpha + \beta)^+,$$

$$\begin{aligned}\alpha + \beta &= \sup \{ \alpha + \lambda \mid \lambda < \beta \} \\ &= \bigcup_{\lambda < \beta} (\alpha + \lambda), \text{ for all limit ordinal } \beta.\end{aligned}$$

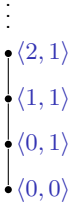
# Ordinal Addition

## Example

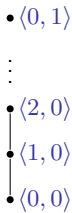
$\omega = 1 + \omega$ ,  $1 + \omega \neq \omega + 1$  and  $\omega + 1 \neq \omega + \omega$ .



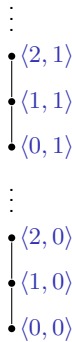
(a)  $\omega$



(b)  $1 + \omega$



(c)  $\omega + 1$



(d)  $\omega + \omega$

## References

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Herbert B. Enderton (1977). Elements of Set Theory. Academic Press (cit. on p. 2).